Principles of communication systems

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Homework #06 Solutions

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Problem 1

a) first let's calculate $\frac{d}{dt}(u(t)\,t)$ using the chain rule

$$\frac{d}{dt}(u(t)\,t) = \frac{du(t)}{dt}\,t + u(t)\frac{dt}{dt} = \delta(t)\cdot t + u(t)$$

$\rightarrow$ claim: $\delta(t)\cdot t = 0$

Proof: from the definition of $\delta(t) \begin{cases} = 0 & , |t| > 0 \\ \neq 0 & , t = 0 \end{cases}$

Therefore $\delta(t)\,t \begin{cases} = 0 & , |t| > 0 \\ = 0 & , t = 0 \end{cases} = 0$

Problem 1

a) Now we can calculate $f(t)$

$$f(t) = \frac{d}{dt} \varphi_{FM}(t) = A\left[w_c + k_f u(t)\right] \cos\left(w_c t + k_f u(t) t\right)$$

$$g(t) = \left| A\left[w_c + k_f u(t)\right] \right|$$ envelope is always positive

b) $$f(t) = -A\left[w_c + k_f \alpha \cos(w_m t)\right] \sin\left(w_c t + \frac{k_f \alpha}{w_m} \sin(w_m t)\right)$$

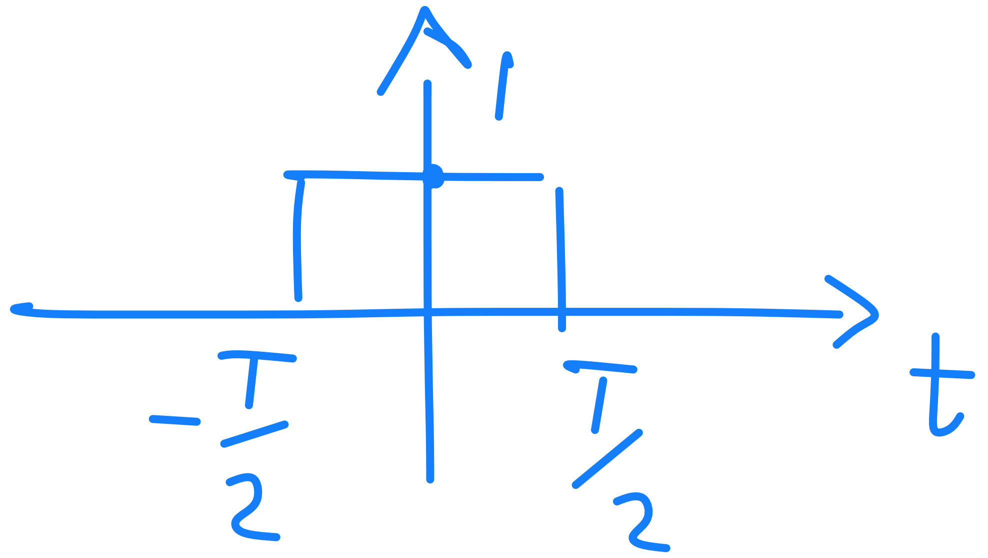
$$g(t) = \left| A\left[w_c + k_f \alpha \cos(w_m t)\right] \right|$$

Problem 2

Nyquist Rate $= 2B$, B: bandwidth of the signal (maximum frequency)

a) First, let's calculate the Fourier transform of $f(t)$.

We know the Fourier transform of a rectangular pulse $p(t) = \Pi\left(\frac{t}{T}\right)$



$$\longleftrightarrow \quad \hat{P}(f) = T\,\frac{\sin(\pi f T)}{\pi f T} = \frac{\sin(\pi T f)}{\pi f}$$

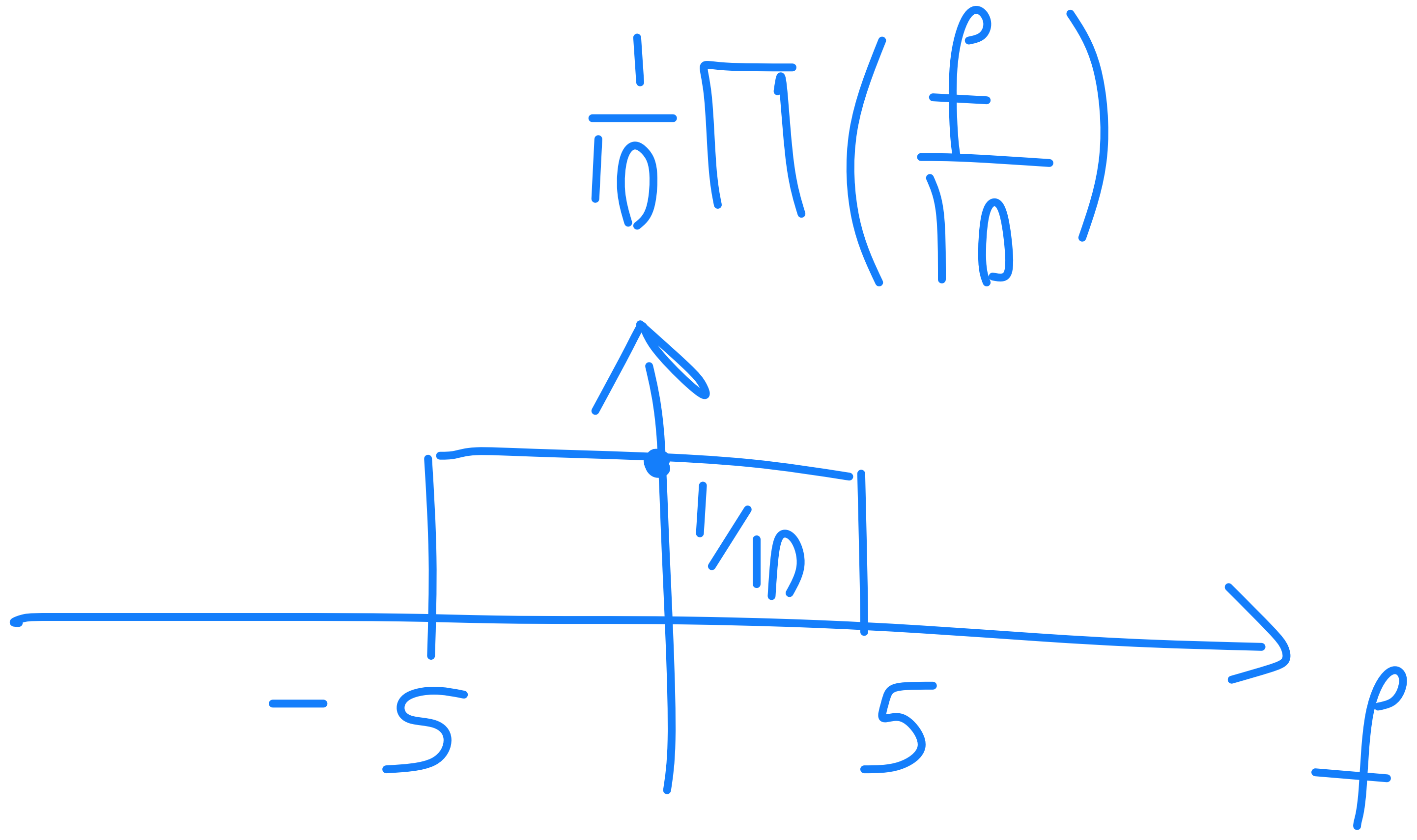
Problem 2

$$f(t) = \frac{\sin(10\pi t)}{10\pi t} = \frac{1}{10}\cdot\frac{\sin(\pi 10 t)}{\pi t}$$

Now, we can write

$$10\,\frac{\sin(\pi 10 t)}{\pi\, 10 t} \xleftrightarrow{\mathcal{F}} \Pi\left(\frac{f}{10}\right)$$

$$f(t) \xleftrightarrow{\mathcal{F}} \frac{1}{10}\,\Pi\left(\frac{f}{10}\right)$$



Therefore, Nyquist Rate = 10 Hz

Problem 2

b) $B = \text{Max frequency} = \max\left(7\times10^3, \dfrac{16000\pi}{2\pi}\right) = 8\times10^3$

Nyquist Rate $= 2B = 16$ kHz

c) $B = 2.3$ kHz

Nyquist Rate $= 2B = 4.6$ kHz